

ResilioSpace: Cloud-Native Self-Adaptive and Self-Healing 2D-to-3D Architectural Visualization Platform

CNAD Project Review-1

29th August, Saturday



Under the Guidance of

Dr. P.Srilatha
Associate Professor
Department of CSE,
KLH CSE Bowrampet Campus



Team Members

MASETTI MANASVINI – 2420030475
DOMA AKSHAYA – 2420030474
NALLAPU SREEJA – 2420030267

Outline of Project Work

- Focus

- Introduction

- Literature Survey

- Challenges

- Research Objectives

- Conclusion & Future Work

- References

- Publications Details

Problem Statement

Traditional architectural design requires manual conversion of 2D floor plans into 3D models, which is time-consuming and prone to errors. Existing systems often lack automatic error recovery, confidence evaluation, and scalable cloud deployment. **ResilioSpace** addresses these challenges by providing a cloud-native platform for automated **2D-to-3D conversion**, **design customization**, **Vastu analysis**, and **self-adaptive/self-healing operation**.

Objectives

- Automate **2D floor-plan to 3D model conversion**.
- Detect **walls, rooms, doors, and windows** using computer vision.
- Provide **interactive 3D visualization and design customization**.
- Enable **Vastu analysis and design evaluation**.
- Support **version comparison** of different designs.
- Implement **confidence-based adaptive processing**.
- Provide **self-healing and automatic failure recovery**.
- Build a **scalable cloud-native application** using Docker and Kubernetes.
- Implement **Helm-based application deployment and management**.
- Automate **CI/CD pipelines** for build, testing, and deployment.
- Integrate **DevSecOps and security scanning** using Trivy.
- Implement **monitoring and observability** using Prometheus and Grafana.
- Apply **GitOps practices** for reliable application deployment.

Literature Survey

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work
Liu et al. [1]	FloorNet: A Unified Framework for Floorplan Reconstruction from 3D Scans	• Uses PointNet and CNN branches to process 3D points, 2D top-view information, and RGB images. • Combines features from multiple inputs for floorplan reconstruction.	• Multi-modal information improves reconstruction quality. • Deep learning can automate floorplan generation.	• Requires RGBD scan data. • Computational complexity can increase for large indoor environments
Lv et al. [2]	Residential Floor Plan Recognition and Reconstruction	• Uses deep segmentation and detection networks to identify room structures. • Uses key-point detection and clustering to estimate room dimensions. • Produces vectorized 3D reconstruction through iterative optimization.	• Automated recognition can improve floor-plan reconstruction accuracy. • Supports vectorized 3D reconstruction from residential floor plans.	• Complex layouts and unusual structures remain challenging. • Further improvement in generalization and reconstruction robustness is possible.
Chen et al. [3]	Floor-SP: Inverse CAD for Floorplans by Sequential Room-Wise Shortest Path	• Uses room segmentation and optimization to reconstruct floorplan	• Optimization-based reconstruction can handle complex	• Primarily designed for RGBD scans rather than simple 2D floor-plan images.

Problem Gap Identification

- Existing methods mainly focus on **2D floor-plan recognition and 3D reconstruction**.
- Limited support for **automatic confidence evaluation and error correction**.
- Lack of **self-adaptive and self-healing mechanisms** in architectural visualization systems.
- Limited integration of **3D customization, Vastu analysis, and version comparison** in one platform.
- Existing solutions do not fully combine these features with **cloud-native deployment, monitoring, CI/CD, and DevSecOps**.
- **ResilioSpace** addresses these gaps through an integrated, scalable, and resilient cloud-native platform.

Innovation, Creativity, and Novelty

- **Automated 2D-to-3D conversion** using computer vision for floor-plan interpretation.
- **Confidence-based adaptive processing** to improve reconstruction accuracy.
- **Self-adaptive and self-healing architecture** using the MAPE-K approach.
- **Interactive 3D design customization** for walls, floors, doors, windows, and materials.
- **Integrated Vastu analysis and design comparison** for better decision-making.
- **Cloud-native deployment** using Docker, Kubernetes, and Helm.
- **DevOps and DevSecOps integration** through CI/CD, security scanning, and monitoring.
- Combines **architectural visualization, intelligent adaptation, and cloud resilience** in a single platform.

Feasibility and Project Plan

Week	Phase	Major Activities / Expected	Output
Week 1	Requirements & Architecture	Finalize ResilioSpace requirements, identify user roles, design cloud-native architecture, define modules, APIs, database schema and workflow.	Requirements, architecture diagram, database design, API/module list
Week 2	Frontend & Backend Development	Develop React.js interface, FastAPI backend, user authentication, floor-plan upload, project management and basic APIs.	Working frontend, backend APIs and upload module
Week 3	2D Processing & 3D Generation	Implement OpenCV-based floor-plan processing, detect walls, rooms, doors and windows, evaluate confidence, and generate interactive 3D models using Three.js.	Floor-plan processing, confidence evaluation and 3D model generation
Week 4	Design & Adaptive Features	Implement 3D customization, Vastu analysis, version comparison, MAPE-K adaptive engine and manual correction for low-confidence results.	Design editor, Vastu analyzer, version comparison and adaptive features
Week 5	Containerization & Kubernetes	Containerize services using Docker, create Kubernetes deployments, configure Helm charts, services, health checks and local deployment using Minikube.	Docker containers, Kubernetes/Helm deployment
		Configure GitHub Actions CI/CD.	